

Null-space energy placement accompanies lower range and INT4 activation error across layers. (a) Whole-activation energy fraction in the groupwise quantizer null space for Hadamard, DuQuant, and PrismQuant $k=\max$ across all 28 Llama-3.2-3B down-input layers. The unlabeled dashed reference is $1/128$. DuQuant is the display label for the existing DuQuant-style diagnostic, not the complete official implementation. (b,c) Paired mean group range and dynamic asymmetric group-128 INT4 activation NMSE for Hadamard, DuQuant, and PrismQuant. The arithmetic means of the 28 per-layer PrismQuant-versus-Hadamard percentage reductions are 25.30% and 40.41%. These are descriptive layer measurements; no seed confidence intervals are inferred. The shared figure-level legend applies to all panels.